

JEE-Main-31-01-2024 (Memory Based)

[EVENING SHIFT]

Physics

Question: The speed of sound in oxygen in STP will be approximately? Given that $R = 8.3$ and $\gamma = 1.4$

Options:

- (a) 330 m/s
- (b) 300 m/s
- (c) 260 m/s
- (d) 350 m/s

Answer: (a)

Question: By what Percentage will the illumination of Lamp Decreases if the current drops by 20%?

Options:

- (a) 20%
- (b) 36%
- (c) 49%
- (d) 60%

Answer: (b)

Question: Mass of the moon is $1/144$ times mass of a planet. Its diameter is $1/16$ the diameter of the planet if the escape velocity of the planet is V then the escape velocity of the moon will be

Options:

- (a) $v/6$
- (b) $v/2$
- (c) $v/3$
- (d) v

Answer: (c)

Question: 3 moles of oxygen gas and 2 moles of argon gas are mixed together. If the total energy of mixture is xRT . Find the value of x .

Options:

- (a) $3 RT / 2$
- (b) $15 RT / 2$
- (c) $7 RT / 2$
- (d) $21 RT / 2$

Answer: (d)

Question: A Nucleus X Having mass number 192 and a second nucleus Y Having radius half of X, then find Mass Number of Nucleus Y

Options:

- (a) 24

- (b) 36
- (c) 42
- (d) 64

Answer: (a)

Question: Force on a 2 kg particle varies with time as, $\vec{F} = 6t\hat{i} - 6t^2\hat{j}$. Find power delivered.

Options:

- (a) $12t^3 - t^5$
- (b) $9t^3 - 6t^5$
- (c) $9t^2 - t^5$
- (d) $5t^3 - 7t^5$

Answer: (b)

Question: A particle is projected at an angle 45° with horizontal with speed u . Find the angular momentum of particle about the point of projection at the time when it reaches maximum height.

Options:

- (a) $mv^3 / 4\sqrt{2}g$
- (b) $3mv^3 / 4\sqrt{2}g$
- (c) $mv^3 / 4g$
- (d) $mv^3 / \sqrt{2}g$

Answer: (a)

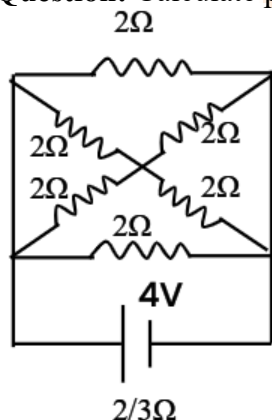
Question: Magnetic flux passing through a loop of resistance 8Ω is given by $\phi = 5t^2 - 3t + 5$. Find current in the loop at $t = 2$ sec.

Options:

- (a) 1 A
- (b) 2 A
- (c) 3 A
- (d) 4 A

Answer: (b)

Question: Calculate power delivered in the circuit



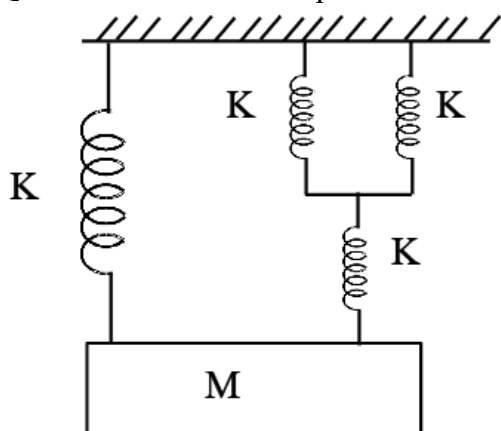
Options:

- (a) 5
- (b) 8
- (c) 7

(d) 4

Answer: (d)

Question: Find the time period of oscillation of the block



Options:

(a) $2\pi\sqrt{\left(\frac{3m}{5k}\right)}$

(b) $\pi\sqrt{\left(\frac{3m}{5k}\right)}$

(c) $7\pi\sqrt{\left(\frac{3m}{5k}\right)}$

(d) $3\pi\sqrt{\left(\frac{3m}{5k}\right)}$

Answer: (a)

Question: A point object is placed at 100 cm from a convex spherical refractive surface having radius of curvature 200 cm and refractive index of the refractive surface is 1.5. Find image distance

Options:

(a) 100 cm

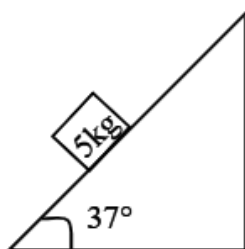
(b) 200 cm

(c) 300 cm

(d) 400 cm

Answer: (c)

Question: Find the difference between minimum force required to prevent the block from sliding down and minimum force required to just push it up the plane. $\mu = 0.1$



Options:

- (a) 10 N
- (b) 8 N
- (c) 16 N
- (d) 4 N

Answer: (b)

Question: S1 : EM Waves Possess Energy

S2 : When EM Waves Strike a Surface, they apply pressure on it

Options:

- (a) S1 : true, S2 : False
- (b) S2 : true, S1 : False
- (c) S1 & S2: true
- (d) S1 & S2: false

Answer: (c)

Question: The force between two charged particles separated by distance r when placed in air is F . if these charges are immersed in a medium of dielectric constant $k = 5$ then find the separation between them to keep the forces same

Options:

- (a) $r/\sqrt{2}$
- (b) $r/\sqrt{3}$
- (c) $r/\sqrt{4}$
- (d) $r/\sqrt{5}$

Answer: (d)

Question: Two vectors of equal magnitude A are inclined at an angle θ with each other. Find the magnitude of resultant vector

Options:

- (a) $2A \cos (\theta)$
- (b) $2A \cos (\theta / 2)$
- (c) $A \cos (\theta / 2)$
- (d) $4A \cos^2 (\theta / 2)$

Answer: (b)

Question: Length of a pendulum is 20 cm and error in its measurement is 2 mm. If it completes 50 oscillations in 40 seconds and time was measured by a watch of resolution 1 sec. Find percentage error in calculation of gravitational acceleration.

Options:

- (a) 3%
- (b) 4%
- (c) 6%
- (d) 7%

Answer: (c)

Question: A nucleus has mass number A_1 for and volume V_1 , another nucleus has mass number A_2 and d volume V_2 . If relation between mass number is $A_2 = 4A_1$, then $V_2 = \underline{\hspace{1cm}} V_1$ [numerical]

Answer: (4)

Question: $E = \frac{B - x^2}{At}$

E → energy

x → length

t → time

then dimensional Formula of AB

Options:

(a) $[M^{-1} L^2 T]$

(b) $[M^1 L^{-2} T]$

(c) $[M^{-1} L^2 T^{-2}]$

(d) $[M^{-1} L^2 T^{-1}]$

Answer: (a)

Question: Frequency is halved and intensity become doubled, then number of photo electrons will be? The original $f = 1.5 f_0$

Options:

(a) Quadrupled

(b) Doubled

(c) Halved

(d) Zero

Answer: (d)

Question: Unpolarised light incident on transparent glass at incident angle 60° . If reflected ray is completely polarized, then angle of refraction is

Options:

(a) 45°

(b) 60°

(c) 30°

(d) 37°

Answer: (c)

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[EVENING SHIFT]

Chemistry

Question: Which of the following is least ionic ?

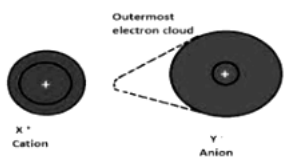
Options:

- (a) BaCl_2
- (b) KCl
- (c) AgCl
- (d) CoCl_2

Answer: (c)

Solution:

Due to Fajans Rule



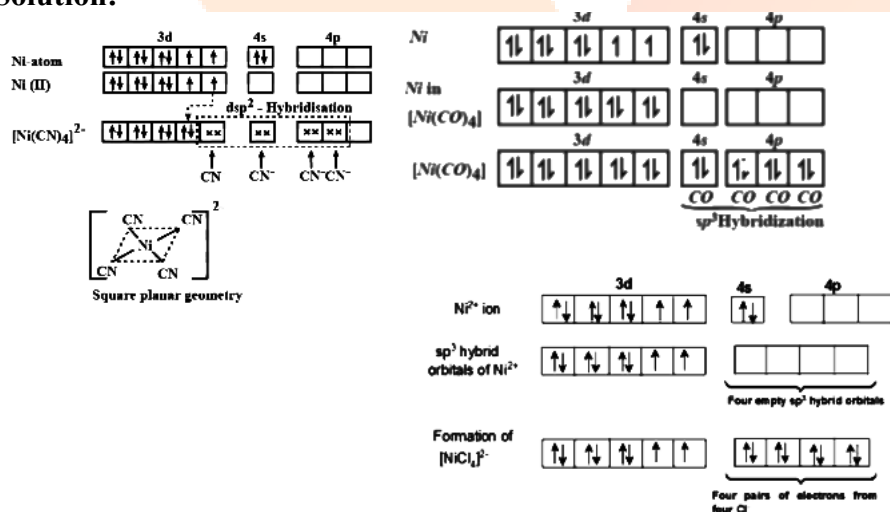
Question: Amongst $\text{Ni}(\text{CO})_4$, $[\text{Ni}(\text{CN})_4]^{2-}$ and NiCl_4^{2-} :

Options:

- (a) $\text{Ni}(\text{CO})_4$ and NiCl_4^{2-} are diamagnetic; and $[\text{Ni}(\text{CN})_4]^{2-}$ is paramagnetic.
- (b) $[\text{Ni}(\text{CN})_4]^{2-}$ and NiCl_4^{2-} are diamagnetic; and $\text{Ni}(\text{CO})_4$ is paramagnetic.
- (c) $\text{Ni}(\text{CO})_4$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic; and NiCl_4^{2-} is paramagnetic
- (d) $\text{Ni}(\text{CO})_4$ is diamagnetic; $[\text{Ni}(\text{CN})_4]^{2-}$ and NiCl_4^{2-} is paramagnetic

Answer: (c)

Solution:



Question: **Statement I:** S_8 disproportionates into $\text{H}_2\text{S}_2\text{O}_3$ and S^{2-} in alkaline medium

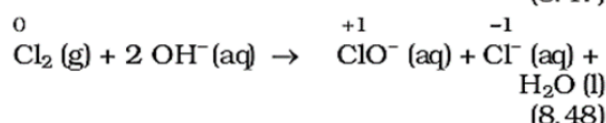
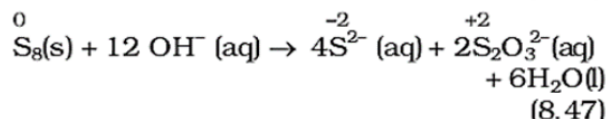
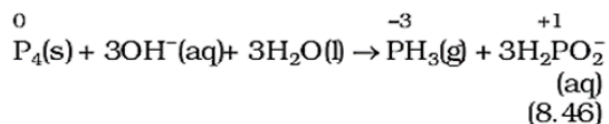
Statement II: ClO_4^- undergoes disproportionation in acidic medium

Options:

- (a) Statement I is correct but Statement II is incorrect
 (b) Statement I is incorrect but Statement II is incorrect
 (c) Both Statement I and Statement II are correct
 (d) Both Statement I and Statement II are incorrect

Answer: (c)

Solution: Phosphorous, sulphur and chlorine undergo disproportionation in the alkaline medium as shown below



Question: Value of all quantum number for outermost electron of Potassium

Options:

- (a) $n = 2, l = 0, m = 0, s = +1/2$
 (b) $n = 4, l = 0, m = 0, s = +1/2$
 (c) $n = 3, l = 1, m = 0, s = +1/2$
 (d) $n = 4, l = 2, m = -1, s = +1/2$

Answer: (b)

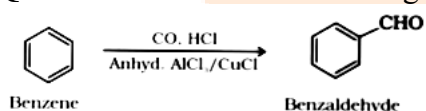
Solution: Electronic configuration of potassium $\text{K} \Rightarrow [\text{Ar}]4s^1$

The outermost orbital is 4s

Hence the 4 quantum numbers are -

$n = 4, l = 0, m = 0, s = +1/2$

Question: What is name of the given reaction :

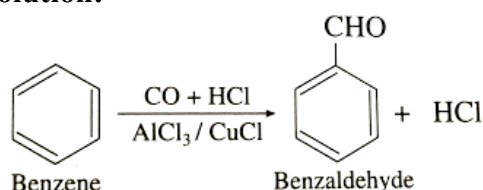


Options:

- (a) Kolbe's Synthesis
 (b) Gatterman Koch
 (c) Sandmeyer
 (d) Gatterman

Answer: (b)

Solution:



Question: Statement-I: Aniline does not undergo Friedel - Crafts reaction (alkylation and acetylation)

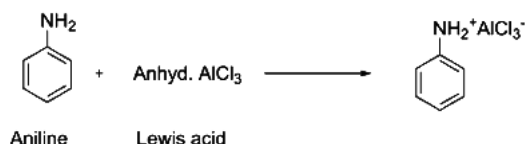
Statement-II: This is due to salt formation with aluminium chloride

Options:

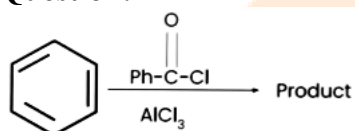
- (a) Statement I is correct but Statement II is incorrect
- (b) Statement I is incorrect but Statement II is incorrect
- (c) Both Statement I and Statement II are correct
- (d) Both Statement I and Statement II are incorrect

Answer: (c)

Solution: Aniline forms salt with the Lewis acid catalyst i.e., AlCl_3 , which is used in Friedel-Crafts reaction.



Question:

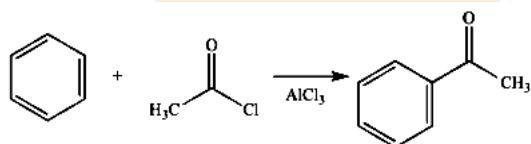


Options:

- (a) Benzophenone
- (b) Acetophenone
- (c) Acid Chloride
- (d) Chloro benzene

Answer: (a)

Solution:



Question: Find the Molarity of 1 L solution of H_2SO_4 given density 1.54 g / cc and have purity 70%

Options:

- (a) 13.2 M
- (b) 12.2 M
- (c) 10.2 M
- (d) 9.2 M

Answer: (a)

Solution:

Mass % = 70 % i.e. 70 g of H_2SO_4 is present in 100 g of the solution.

Density = $1.84 \text{ g cc}^{-1} = 1.84 \text{ g mL}^{-1}$

Hence, volume = $\frac{100}{1.84} \text{ mL} = \frac{1}{18.4} \text{ L} = 0.054$

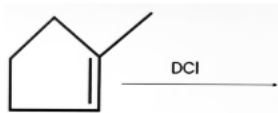
Given mass of $\text{H}_2\text{SO}_4 = 70$

Number of moles of $\text{H}_2\text{SO}_4 = \frac{70 \text{ g}}{98} = 0.7142$

Molarity = $\frac{\text{moles}}{\text{volume of the solution in L}}$

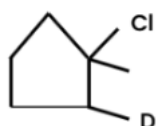
$$\text{Molarity} = \frac{0.17142}{0.059} = 13.22 \text{ M}$$

Question:

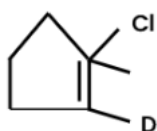


Options:

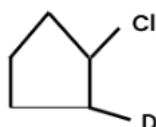
(a)



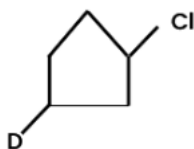
(b)



(c)



(d)



Answer: (a)

Solution:



Question: There is some fragrance oil found in flowers which are water insoluble but can be mixed water in vapour phase. What is the best method to extract this oil compound ?

Options:

- (a) Distillation in reduced pressure.
- (b) Steam distillation
- (c) Distillation
- (d) Crystallization

Answer: (b)

Solution: The component is miscible with water vapour in vapour phase so steam distillation will be most suitable method for extraction.

Question: Correct Order of Electrophilic Aromatic Substitution Reaction Toluene, Benzene, Chloroform and Nitrobenzene

Options:

- (a) $1 > 2 > 3 > 4$
- (b) $4 > 3 > 2 > 1$
- (c) $2 > 1 > 3 > 4$
- (d) $2 > 3 > 1 > 4$

Answer: (c)

Solution:

$-\text{CH}_3$ - Electron donating group

$-\text{Cl}$ - Electron withdrawing group ($-\text{I}$ dominates over $+\text{R}$ effect)

$-\text{NO}_2$ - Strongly electron withdrawing group

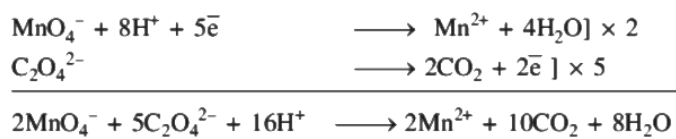
Question: For the redox reaction, $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$, the correct coefficients of the reactants for the balanced equation are _____.

Options:

- (a) $\text{MnO}_4^- = 2, \text{C}_2\text{O}_4^{2-} = 16, \text{H}^+ = 5$
- (b) $\text{MnO}_4^- = 16, \text{C}_2\text{O}_4^{2-} = 5, \text{H}^+ = 2$
- (c) $\text{MnO}_4^- = 5, \text{C}_2\text{O}_4^{2-} = 16, \text{H}^+ = 2$
- (d) $\text{MnO}_4^- = 2, \text{C}_2\text{O}_4^{2-} = 5, \text{H}^+ = 16$

Answer: (d)

Solution:



Question: For a diatomic molecule having bond length $1 \text{ \AA}$, $\mu = 1.25 \text{ Debye}$ Find q .

$1 \text{ D} = 10^{-18} \text{ esu cm}$

Options:

- (a) 2.2×10^{-10}
- (b) 1.2×10^{-10}
- (c) 1.8×10^{-10}
- (d) 4.8×10^{-10}

Answer: (b)

Solution:

Dipole moment (μ) = charge (q) $\times$ distance (d)

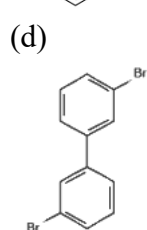
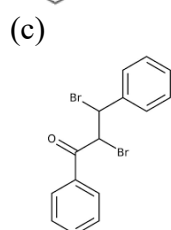
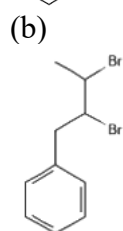
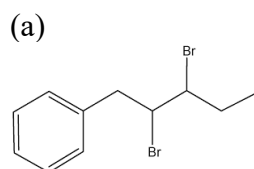
$$\mu = q \times d$$

$$1.2 \times 10^{-18} = q \times 10^{-8} \text{ cm}$$

$$q = \frac{1.2 \times 10^{-18}}{10^{-8}} = 1.2 \times 10^{-10}$$

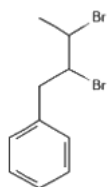
Question: Correct structure of 2, 3 dibromo-1 phenyl butane

Options:



Answer: (b)

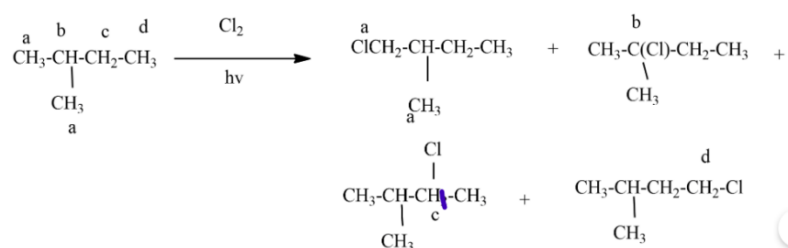
Solution:



Question: Number of isomeric products formed by monochlorination of 2 - methyl butane in presence of sunlight is

Answer: 6

Solution:



Question: From the vitamins A, B - 1, B - 6, B - 12, C, D, and K, the number of vitamins that can be stored in our body is

Answer: 3

Solution: Water soluble vitamins can't be stored in our body.

B group vitamins (B_1 , B_2 , B_6 , B_{12} , etc.) and vitamin C are water soluble vitamins.

Question: If 5 moles of an ideal gas expands from 10 L to a volume of 100 L at 300 K under isothermal and reversible condition then work, W , is $-x$ J. The value of x is ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

Answer: 28720

Solution:

Work done in an isothermal irreversible expansion is given by the formula :

$$W = -nRT \ln \left(\frac{V_f}{V_i} \right)$$

$$= -2.303 \times 5 \times 8.314 \times 300 \log 10$$

$$= -28720 \text{ J}$$

Question: Half life of a first order reaction is 36 hr. Find out time (in hour) required for concentration of reactant to get reduced by 90%.

Answer: 118.8

Solution:

For a first order reaction,

$$k = \frac{2.303}{t} \log \frac{a}{a-x}$$

$$t = \frac{2.303 \times t_{1/2}}{0.693} \log \frac{100}{(100-90)} \quad [\because k = \frac{t_{1/2}}{0.693}]$$

$$= \frac{2.303 \times t_{1/2}}{0.693} \log 10$$

$$= 3.3 t_{1/2}$$

$$= 3.3 \text{ times that of half-life.}$$

$$= 3.3 \times 36 \text{ hrs}$$

$$= 118.8 \text{ hrs}$$

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Mathematics

Question: Let $f : R \rightarrow (0, \infty)$ be increasing function such that $\lim_{x \rightarrow \infty} \frac{f(7x)}{f(x)} = 1$ then

$\lim_{x \rightarrow \infty} \left\{ \frac{f(5x)}{f(x)} - 1 \right\}$ is equal to

Options:

- (a) 0
- (b) 4
- (c) 1
- (d) $\frac{4}{5}$

Answer: (a)

Solution:

$$\lim_{x \rightarrow \infty} \frac{f(7x)}{f(x)} = 1$$

$$f(x) \leq f(5x) \leq f(7x)$$

$$1 \leq \frac{f(5x)}{f(x)} \leq \frac{f(7x)}{f(x)}$$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{f(5x)}{f(x)} - 1 \right) = 0$$

Question: If $z_1^3 + z_2^3 = 20 + 15i$ and $z_1 + z_2 = 5$ then $|z_1^4 + z_2^4|$ is equal to:

Options:

- (a) $15\sqrt{5}$
- (b) 75
- (c) $30\sqrt{3}$
- (d) $25\sqrt{3}$

Answer: (b)

Solution:

$$|z_1^4 + z_2^4| \text{ can be written as } (z_1^2 + z_2^2)^2 - 2z_1^2 z_2^2$$

$$z_1 + z_2 = 5$$

Cubing on both sides

$$(z_1 + z_2)^3 = (5)^3$$

$$20 + 15i + 3z_1 z_2 (5) = 125$$

$$z_1 z_2 = \frac{105 - 15i}{15}$$

$$z_1 z_2 = 7 - i$$

$$\begin{aligned} |z_1^4 + z_2^4| &= (z_1^2 + z_2^2)^2 - 2z_1^2 z_2^2 \\ &= [(z_1 + z_2)^2 - 2z_1 z_2]^2 - 2(7 - i)^2 \\ &= [25 - 2(7 - i)]^2 - 2(7 - i)^2 \end{aligned}$$

By simplifying

$$|z_1^4 + z_2^4| = 75$$

Question: $a = \sin^{-1}(\sin(5))$ and $b = \cos(\cos(5))$ then $a^2 + b^2 =$

Options:

(a) $8\pi^2 - 40\pi + 50$

(b) $4\pi^2 + 25$

(c) $8\pi^2 - 50$

(d) $8\pi^2 + 40\pi + 50$

Answer: (a)

Solution:

$$a = \sin^{-1}(\sin 5)$$

$$a = \sin^{-1} \sin(2\pi + 5 - 2\pi) \quad (\because 5 - 2\pi \approx 1.78)$$

$$a = 5 - 2\pi$$

$$b = \cos^{-1}(\cos 5)$$

$$= \cos^{-1}(\cos(-5))$$

$$= \cos^{-1}(\cos(-2\pi + 2\pi - 5))$$

$$= \cos^{-1} \cos(2\pi - 5)$$

$$b = 2\pi - 5$$

$$a^2 + b^2 = 2(25 + 4\pi^2 - 10\pi)$$

$$= 8\pi^2 - 40\pi + 50$$

Question: The number of ways distribute the 21 identical apples to these children so that each child gets at least 2 apples is?

Answer: 136.00

Solution:

$$x + y + z = 21$$

$$(x' + 2)(y' + 2) + (2' + 2) = 21$$

$$x' + y' + z' = 15$$

$$\text{Number of ways} = {}^{(15+2)}C_2 \Rightarrow {}^{17}C_2 \Rightarrow \frac{17 \times 16}{9} = 136$$

Question: A coin is biased so that a head is twice as likely occurs as a tail if the coin is tossed 3 times, then the probability of getting two tails and one head is

Options:

- (a) $\frac{1}{9}$
- (b) $\frac{2}{9}$
- (c) $\frac{2}{27}$
- (d) $\frac{1}{27}$

Answer: (b)

Solution:

$$P(T) = \frac{1}{3}, P(H) = \frac{2}{3}$$

$$\text{Probability} = {}^3C_2 \left(\frac{1}{3}\right)^2 \cdot \frac{2}{3} = 3 \cdot \frac{1}{9} \cdot \frac{2}{3} = \frac{2}{9}$$

Question: The number of solution of equation $e^{\sin x} - 2e^{-\sin x} = 2$ is

Options:

- (a) more than 2
- (b) 2
- (c) 1
- (d) 0

Answer: (d)

Solution:

$$e^{\sin x} - 2e^{-\sin x} = 2$$

$$\Rightarrow t - \frac{2}{t} = 2, t = e^{\sin x} \in \left[\frac{1}{e}, e\right]$$

$$\Rightarrow t^2 - 2t - 2 = 0$$

$$\Rightarrow t = \frac{2 \pm \sqrt{4+8}}{2} = 1 \pm \sqrt{3} \in \left[\frac{1}{e}, e\right]$$

No Solution

Question: The area of region enclosed by parabola $y = 4x - x^2$ and $3y = (x - 4)^2$

Answer: 6.00

Solution:

$$y = 4x - x^2$$

$$x^2 - 4x + 4 = y + 4$$

$$(x - 2)^2 = -(y - 4)$$

Point of intersection

$$12x - 3x^2 = x^2 - 8x + 16$$

$$= 4x^2 - 20x + 16$$

$$x^2 - 5x + 4$$

$$(x - 1)(x - 4)$$

$$\int_1^4 4x - x^2 - \frac{(x^2 - 8x + 16)}{3}$$

$$- \frac{4}{3} \left(\frac{x^3}{6} - \frac{5x^2}{2} + 4x \right) \Bigg|_1^4 = - \frac{4}{3} \left[\frac{64}{3} - \frac{80}{2} + 16 - \frac{1}{2} + \frac{5}{2} - 4 \right] = 6$$

Question: If 2nd, 8th, 44th terms of A.P. are 1st, 2nd and 2rd terms respectively of G.P. and first term of A.P. is 1, then the sum of first 20 terms of A.P. is:

Options:

- (a) 970
- (b) 916
- (c) 980
- (d) 990

Answer: (a)

Solution:

A.P. - 1, d

G.P. - a, r

$$1 + d = a$$

$$1 + 7d = ar$$

$$1 + 43d = ar^2$$

$$(ar^2)a = (ar^2)^2$$

$$(1 + 43d)(1 + d) = (1 + 7d)^2$$

$$1 + 44d + 43d^2 = 1 + 14d + 49d^2$$

$$0 - 6d^2 - 30d$$

$$d = 0, d = 5$$

$$S_{20} = \frac{70}{2}(2 + 19.5)$$

$$= 10(97)$$

$$= 970$$

Question: $\lim_{x \rightarrow 0} \frac{ax^2e^x - b \log_e(1+x) + cxe^{-x}}{x^2 \sin x} = 1$ then $16(a^2 + b^2 + c^2) =$

Answer: 81.00

Solution:

$$\lim_{x \rightarrow 0} \frac{ax^2 \left(1 + \frac{x}{1!} + \frac{x^2}{2!}\right) - b \left(x - \frac{x^2}{2} + \frac{x^3}{3}\right) + cx \left(1 - \frac{x}{1!} + \frac{x^2}{2!}\right)}{x^2 x} = 1$$

$$\frac{ax^2 + ax^3 - bx + \frac{bx^2}{2} - \frac{bx^3}{3} + cx - cx^2 + \frac{cx^3}{2}}{x^3} = 1$$

$$-b + c = 0 \quad a + \frac{b}{2} - c = 0$$

$$c = b \quad a = \frac{c}{2}$$

$$a - \frac{b}{3} + \frac{c}{2} = 1$$

$$\frac{c}{2} - \frac{c}{3} + \frac{c}{2} = 1$$

$$\frac{2c}{3} = 1$$

$$c = \frac{3}{2}, b = \frac{3}{2}, a = \frac{3}{4}$$

$$16(a^2 + b^2 + c^2)$$

$$16 \times \left(\frac{81}{16}\right) = 81$$

Question: A is a square matrix of order 3 is $\det(A) = 2$ then find the value of

$$\underbrace{\det(\text{adj } \text{adj } \dots \text{adj } A)}_{2024 \text{ times}}.$$

Answer:

Solution:

$$|\text{adj}(\text{adj}(\text{adj } A))| = |A|^{(n-1)^{2024}}$$

$$\underbrace{\det(\text{adj } \text{adj } \dots \text{adj } A)}_{2024 \text{ times}}$$

$$= (2)^{(3-1)^{2024}}$$

$$= (2)^{(2)^{2024}}$$

Question: Let the mean and variance of 6 observations $a, b, 68, 44, 48, 60$ be 55 and 194 respectively. If $a > b$ then $a + 3b$ is

Options:

- (a) 180
- (b) 190
- (c) 210
- (d) 200

Answer: (a)

Solution:

$a, b, 68, 44, 48, 60$

$$\frac{a + b + 220}{6} = 55$$

$$a + b = 110$$

$$194 = \frac{a^2 + b^2 + 12464}{6} - (55)^2$$

$$194 = \frac{a^2 + b^2}{6} + 2077\frac{1}{3} - 3075$$

$$19314 - 12464 = a^2 + b^2$$

$$a^2 + b^2 = 6850$$

$$a^2 + b^2 + 2ab = 12100$$

$$2ab = 5750$$

$$ab = 2675$$

$$a = 75$$

$$b = 35$$

$$a + 3b = 180$$